

PHY 115L

The Gaussian Wave Packet

Michael Mulhearn

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1 Introduction

For this assignment, we will animate solutions to the 1-D time-dependent Schrödinger Equation (TDSE):

$$i\hbar \frac{\partial \Psi}{\partial t} = H\Psi(x, t) \quad (1)$$

where H is the Hamiltonian operator:

$$H = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)$$

The energy eigenstates of the Hamiltonian:

$$H |n\rangle = E_n |n\rangle$$

are a basis for the Hilbert space and have a well understood time-dependence:

$$|n(t)\rangle = \exp\left(-\frac{iE_n t}{\hbar}\right) |n\rangle$$

So once we know the coefficients of the energy eigenstates for the wave function at $t = 0$:

$$c_n = \langle n | \Psi(0) \rangle$$

then we have the time-dependent solution as well:

$$|\Psi(t)\rangle = \sum_n c_n \exp\left(-\frac{iE_n t}{\hbar}\right) |n\rangle \quad (2)$$

2 System of Units

As ever, we will employ a numerical system with a characteristic length scale a . In numerical analysis, it is more natural to use the wave number k , related to the momentum p , by:

$$p = \hbar k$$

In our system of units k is measured in units of $1/a$. Equivalently, we can consider momentum to be measured in units of:

$$p_0 = \hbar/a$$

Energy will be measured in units of:

$$E_0 \equiv \frac{\hbar^2}{2ma^2} \quad (3)$$

As always, quantities like $\hbar$, a , p_0 and E_0 should not appear in your program. We keep track of them explicitly for analytic calculations, where dimensional analysis is useful, but we set them to one in our computer programs.

3 Time-Evolution of a Free Particle Wave Packet

For a free particle, the energy eigenstates are the momentum eigenstates, which are continuous. Therefore, the sum in Equation 2 is replaced by an integral:

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) \exp \left[ik \left(x - \frac{\hbar k}{2m} t \right) \right] dk \quad (4)$$

where the discrete coefficients have been replaced by the continuous Fourier transform of the initial state:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Psi(x, 0) e^{-ikx} dx \quad (5)$$

We will consider the initial state to be:

$$\Psi(x, 0) = N \exp \left(-\frac{x^2}{4\alpha^2} \right) \exp(i\beta x) \quad (6)$$

with normalization factor:

$$N = \left(\frac{1}{2\pi\alpha^2} \right)^{\frac{1}{4}}$$

Notice that:

$$|\Psi(x, 0)|^2 = \frac{1}{\sqrt{2\pi\alpha}} \exp \left(-\frac{x^2}{2\alpha^2} \right)$$

which is a Gaussian distribution with width α . The Fourier transform of this initial state is:

$$\begin{aligned} \phi(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-ikx} \Psi(x, 0) dx \\ &= \frac{N}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp(-X) dx \end{aligned}$$

where:

$$\begin{aligned} X &= \frac{1}{4} \left(\frac{x}{\alpha} \right)^2 + i(k - \beta) x \\ &= \frac{1}{4} \left[\frac{x}{\alpha} + i2\alpha(k - \beta) \right]^2 + \alpha^2(k - \beta)^2 \end{aligned}$$

and we have “completed the square”. To compute the integral, we make the substitution:

$$u = \frac{1}{2} \left[\frac{x}{\alpha} + i2\alpha(k - \beta) \right]$$

so:

$$X = u^2 + \alpha^2(k - \beta)^2$$

and

$$dx = 2\alpha du.$$

Our new integration variable u is complex, and our integral is now a contour integral on the complex plane, but we will get the correct result by simply rotating back to the x -axis, e.g.:

$$u = \pm\infty + i2\alpha(k - \beta) \rightarrow \pm\infty$$

so the integral becomes:

$$\phi(k) = M \exp(-\alpha^2(k - \beta)^2)$$

where

$$M = \frac{2\alpha N}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2} du = \left(\frac{2\alpha^2}{\pi} \right)^{\frac{1}{4}}$$

Notice that:

$$|\phi(k)|^2 = \frac{1}{\sqrt{2\pi} \left(\frac{1}{2\alpha}\right)} \exp\left(-\frac{(k - \beta)^2}{2 \left(\frac{1}{2\alpha}\right)^2}\right)$$

which is a Gaussian distribution with mean β and

$$\sigma_k = \frac{1}{2\alpha}$$

Recalling that $\sigma_x = \alpha$ and $p = \hbar k$ we have:

$$\sigma_x \sigma_p = \frac{\hbar}{2}$$

so the wave packet has minimum uncertainty.

Next we calculate the time-dependent wave function from:

$$\begin{aligned} \Psi(x, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{ikx} \exp\left(\frac{-iEt}{\hbar}\right) dk \\ &= M \int_{-\infty}^{+\infty} \exp(-X) dk \end{aligned}$$

where:

$$X = \alpha^2(k - \beta)^2 - ikx + i\frac{\hbar t}{2m} k^2$$

and we have used:

$$E = \frac{\hbar^2 k^2}{2m}.$$

The algebra that follows turns out much simpler if we first make the substitution:

$$q = k - \beta \implies k = q + \beta$$

So we have:

$$\begin{aligned}
X &= \alpha^2 q^2 - i(q + \beta)x + i\frac{\hbar t}{2m} (q + \beta)^2 \\
&= \left(\alpha^2 + i\frac{\hbar t}{2m} \right) q^2 + i\left(\frac{\hbar \beta t}{m} - x \right) q - i\beta x + i\frac{\hbar \beta^2}{2m} t \\
&= A \left(\frac{q}{\beta} \right)^2 + B \frac{q}{\beta} - i\beta x + i\frac{\hbar \beta^2}{2m} t
\end{aligned}$$

where:

$$\begin{aligned}
A &= n^2 + i\omega t \\
B &= i(2\omega t - \beta x) \\
n &= \alpha\beta \\
\omega &= \frac{\hbar \beta^2}{2m}
\end{aligned}$$

Notice the coefficient of q is pure imaginary, which is the simplification that resulted from our substitution. Completing the square we find:

$$X = A \left[\frac{q}{\beta} + \frac{B}{2A} \right]^2 - \frac{B^2}{4A} - i\beta x + i\omega t$$

and substituting

$$u = \sqrt{A} \left[\frac{q}{\beta} + \frac{B}{2A} \right]$$

we have:

$$X = u^2 - \frac{B^2}{4A} - i\beta x + i\omega t.$$

and:

$$\begin{aligned}
-\frac{B^2}{4A} &= \frac{(2\omega t - \beta x)^2}{4(n^2 + i\omega t)} \\
&= \frac{\beta^2}{4n^2} \frac{(x - \frac{2\omega}{\beta}t)^2}{1 + i(\omega t/n^2)} \\
&= \frac{(x - vt)^2}{4\alpha^2(1 + i(\omega t/n^2))}
\end{aligned}$$

where:

$$v = \frac{2\omega}{\beta} = \frac{\hbar \beta}{m}$$

We also have:

$$-i\beta x + i\omega t = i\beta(x - vt/2)$$

Noting that:

$$dk = dq = \frac{\beta}{\sqrt{A}} du$$

we have:

$$\Psi(x, t) = N(t) \exp\left(-\frac{(x - vt)^2}{4\alpha^2(1 + i(\omega t/n^2))}\right) \exp\left(i\beta\left(x - \frac{vt}{2}\right)\right) \quad (7)$$

where:

$$\begin{aligned} N(t) &= \frac{1}{\sqrt{2\pi}} \left(\frac{2\alpha^2}{\pi}\right)^{\frac{1}{4}} \frac{\beta}{\sqrt{A}} \int_{-\infty}^{\infty} e^{-u^2} du \\ &= \left(\frac{1}{2\pi\alpha^2}\right)^{\frac{1}{4}} \frac{1}{\sqrt{1 + i\omega t/n^2}} \end{aligned}$$

There is a lot of phenomenology contained in Equation 7. Things you can notice:

- The packet starts with initial width α and becomes wider over time via dispersion.
- The packet travels with group velocity $v = \hbar\beta/m$, i.e. the classical velocity of a particle with momentum $\hbar\beta$.
- The carrier waves (complex exponential) have a phase velocity that is one-half the group velocity.

You should arrange your animation so that these features are evident.

4 Animation in Scientific Python

To animate the time dependent solutions, we will use the FuncAnimation feature from matplotlib.animation. To run in a jupyter notebook, you will want to use the matplotlib widget backend for animation:

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.animation import FuncAnimation
%matplotlib widget
```

Remember that the last line is not a comment, it is “line magic” that starts the backend. If you haven’t used the matplotlib widget before, you may have to install it first. For example, under conda:

```
conda install ipympl
```

An example animation is provided on the course website (uncertainty.ipynb). In that example, the animation is implemented like this:

```
# animation parameters
tstep = 1
fps = 20
frame_ms = int(1000 / fps)
second_tau = fps * tstep

# setup the figures
fig, axes = plt.subplots(1, 2, figsize=(16, 9), dpi=1920/32)
xf = np.linspace(-10, 10, 1000)
la, = axes[0].plot([], [], "b-")
lb, = axes[1].plot([], [], "r-")
```

```

axes[0].set_xlim(-10, 10)
axes[1].set_xlim(-10, 10)
axes[0].set_ylim(0, 1.5)
axes[1].set_ylim(0, 1.5)

axes[0].set_xlabel("$x$")
axes[0].set_ylabel("$\\psi(x)$")
axes[1].set_xlabel("$p$")
axes[1].set_ylabel("$\\phi(p)$")

# initialization of plots:
def init():
    la.set_data([], [])
    lb.set_data([], [])
    return la, lb

# update plots once per frame:
def animate(i):
    s = sig(i)
    la.set_data(xf, psi(xf, s))
    lb.set_data(xf, phi(xf, s))
    return la, lb

#run the animation:
anim = FuncAnimation(
    fig,
    animate,
    frames=200,
    init_func=init,
    interval=frame_ms,
    blit=False,
    repeat=True,
)

plt.tight_layout()
plt.show()

```

The main features are the initial setup of animation parameters and setting up the (empty) plots. The init function is called at the start of the animation, but does little in this example. In the animate function, which is called once per frame, the plots are updated with the data for that frame. Last, the animation is called via FuncAnimation.

5 Homework Problems

Problem 1: Get the example program (uncertainty.ipynb) running on your system successfully. You have explicit permission to use the internet, AI, knowledgeable colleagues, etc, in order to accomplish this. Please submit your working code (even if unchanged) for this problem, and include some brief comments about the installation steps that were needed. Hopefully, the example code will just work out of the box for you, in which case this is a very easy problem!

Problem 2: Animate the time evolution of a stationary wave packet ($\beta = 0$) Show the magnitude as a black line, and real and imaginary components as blue and red lines. Use the parameterization from our derivation!

Problem 3: Animate the time evolution of a travelling wave packet. Use the parameterization from our derivation.